

Question 1

Given,

$$E(n, y) = 2.5ny - 0.04n^3 - 0.02y^3$$

$$C(n, y) = 100n + 5y^2 + 3y^2$$

a) Find the partial derivative of $E(n, y)$ w.r.t. n and y

Soln:-

$$E(n, y) = 2.5ny - 0.04n^3 - 0.02y^3 \dots (i)$$

Differentiating eqⁿ(i) w.r.t. n we get,

$$\frac{\partial E}{\partial n} = \frac{\partial (2.5ny)}{\partial n} - \frac{\partial (0.04n^3)}{\partial n} - \frac{\partial (0.02y^3)}{\partial n}$$

$$\frac{\partial E}{\partial n} = 2.5y - 0.04 \cdot 3n^2 - 0$$

$$\frac{\partial E}{\partial n} = 2.5y - 0.12n^2$$

Differentiating eqⁿ(i) w.r.t. y we get,

$$\frac{\partial E}{\partial y} = \frac{\partial (2.5ny)}{\partial y} - \frac{\partial (0.04n^3)}{\partial y} - \frac{\partial (0.02y^3)}{\partial y}$$

$$\frac{\partial E}{\partial y} = 2.5n - 0 - 0.02 \cdot 3y^2$$

$$\frac{\partial E}{\partial y} = 2.5n - 0.06y^2$$

b) Use the concept of critical points to determine the dimensions (length and width) of the panel that maximize energy output within the given constraints.

Soln:-

To determine the dimensions that maximize energy output,

At a critical point,

$$E_x = 0, E_y = 0$$

Therefore,

$$2.5y - 0.12x^2 = 0$$

$$\text{or } 2.5y = 0.12x^2$$

$$\text{or } y = 0.048x^2$$

Also,

$$2.5x - 0.06y^2 = 0$$

$$x = \frac{0.06y^2}{2.5}$$

$$x = 0.024y^2$$

Substitute $x = 0.024y^2$ into $y = 0.048x^2$

$$\text{or } y = 0.048(0.024y^2)^2$$

$$\text{or } y = 0.048(0.000576)y^4$$

$$\text{or } y = 0.000027648y^4$$

For $y \neq 0$

$$\text{or } 1 = 0.000027648y^3$$

$$y^3 = 36169$$

$$\therefore y = 33.1$$

This exceeds the allowed width $y \leq 3$

Hence, there is no interior critical point inside the feasible region, so the maximum must occur on the boundary.

Evaluate $E(n, y)$ at the corner points

Corner $(0, 0)$

$$E(0, 0) = 0$$

Corner $(5, 0)$

$$E(5, 0) = 0 - 0.04(125) = -5$$

Corner $(0, 3)$

$$E(0, 3) = -0.02(27) = -0.54$$

Corner $(5, 3)$

$$\begin{aligned} E(5, 3) &= 2.5(5)(3) - 0.04(125) - 0.02(27) \\ &= 37.5 - 5 - 0.54 \\ &= 31.96 \end{aligned}$$

To check Boundaries

Boundary at $n = 5$

$$E(5, y) = 12.5y - 5 - 0.02y^3$$

$$\frac{dE}{dy} = 12.5 - 0.06y^2$$

For $0 \leq y \leq 3$,

$$12.5 - 0.06(9) = 11.96 > 0$$

So, E increases throughout the interval, giving maximum at $y = 3$.

Boundary at $y = 3$

$$E(n, 3) = 7.5n - 0.04n^3 - 0.54$$

$$\frac{dE}{dn} = 7.5 - 0.12n^2$$

$$\text{At } n = 5, \quad 7.5 - 0.12(5)^2 = 4.5 > 0$$

$$7.5 - 3 = 4.5 > 0$$

Thus E increases on $[0, 5]$ which gives maximum at $n = 5$.

Therefore,

$n = 5m$ and $y = 3m$ maximize the energy output.

c) Calculate the maximum attainable energy output and the corresponding production cost using the optimal dimensions.

Soln:-

The maximum energy at $E(5, 3)$ is 31.96. So, $E_{\max} = 31.96 \text{ MW}$.

The production cost is given in the function

$$C(n, y) = 100n + 8y^2$$

Substituting $n = 5$ and $y = 3$

$$C(5, 3) = 100(5) + 8(3)^2$$

$$= 500 + 72$$

$$= 572$$

$$\therefore C = 572$$

Question 2

Volume of the solid of R is given by

$$y = x^2$$

and

$$y = 4 \text{ with surface } z = 6 - x^2$$

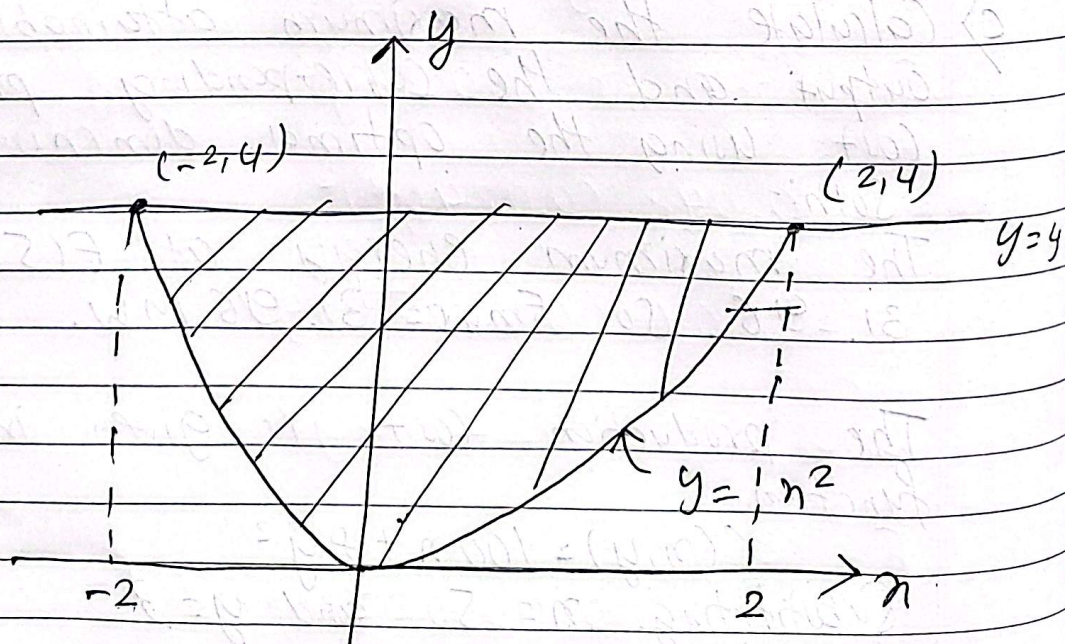
a) Sketch the region R in the xy -plane

To find the intersection point

$$x^2 = 4$$

$$x = \pm 2$$

Thus region R lies between the lower curve $y = x^2$ and upper curve $y = 4$ for $-2 \leq x \leq 2$



b) Set up the double integral for computing the volume of the solid.

Soln:-

Volume under the surface $z = 6 - x^2$ over the region R :

$$V = \iint_R (6 - x^2) dA$$

Using the vertical strips:

$$-2 \leq x \leq 2, \quad x^2 \leq y \leq 4$$

So,

The integral becomes,

$$V = \int_{-2}^2 \int_{x^2}^4 (6 - x^2) dy dx$$

c) Evaluate the double integral to find the volume.

First integrate w.r.t. y :

$$V = \int_{-2}^2 (6 - x^2)(4 - x^2) dx$$

Expand:

$$(6 - x^2)(4 - x^2) = 24 - 10x^2 + x^4$$

Thus,

$$V = \int_{-2}^2 (24 - 10x^2 + x^4) dx$$

Since the integrand is even:

$$V = 2 \int_0^2 (24 - 10x^2 + x^4) dx$$

Substitute $n = 2$:

$$V = 2 \left(48 - \frac{80}{3} + \frac{32}{5} \right)$$

Using the denominator 15:

$$\frac{48}{1} = \frac{720}{15}, \quad \frac{80}{3} = \frac{400}{15}, \quad \frac{32}{5} = \frac{96}{15}$$

$$V = 2 \left(\frac{720 - 400 + 96}{15} \right)$$

$$V = 2 \left(\frac{416}{15} \right)$$

$$V = \frac{832}{15}$$

$$\therefore V = 55.47 \text{ cubic units}$$